NATIONAL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
END-SEMESTER EXAMINATION: DEEP LEARNING & ARTIFICIAL INTELLIGENCE
Course Code: CS702 | Max Marks: 50 | Time Allowed: 2 Hours

INSTRUCTIONS: Answer all questions. Each question carries marks as indicated.

Q1. Define the terms 'vanishing gradient' and 'exploding gradient' in deep neural networks. [5 Marks]

Q2. Explain how the self-attention mechanism in the Transformer architecture computes relevance weights between input tokens. [5 Marks]

Q3. Calculate the output dimensions of a feature map after passing an input image of size 64x64x3 through a convolutional layer with 16 filters of size 5x5, stride 2, and padding 1. [10 Marks]

Q4. Differentiate between WordPiece tokenization, Byte-Pair Encoding (BPE), and simple whitespace tokenization. Compare their handling of out-of-vocabulary words. [10 Marks]

Q5. Critique the decision to use accuracy versus macro-F1 score when evaluating a multi-class model on an imbalanced classification dataset. Justify your argument. [10 Marks]

Q6. Design an end-to-end deep learning architecture to automatically classify semester examination questions into the six cognitive tiers of Bloom's Taxonomy. Formulate the loss function and evaluation strategy. [10 Marks]